MILITARY SPECIFICATION

SHIMS, MOLDED, FILLED RESIN COMPOUND AND SHEET PREPREG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for filled, resinous, thermosetting compound and sheet prepreg to fabricate shims for use during assembly of aircraft components and parts.

1.2 Classification. The material shall be classified in the three types as specified:

Type I - A two component material (base and accelerator) which, after mixing and application of heat (350°F), will cure to a hard, strong resin.

Type II - A precatalyzed (B-staged) sheet material which will cure to a hard, strong resin with the application of heat.

Type III - A precatalyzed (B-staged) material supplied frozen in a tube from which it can be extruded and cured to a hard strong resin with the application of heat (200°F).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-S-809	Sulfuric Acid, Technical
O-S-595	Sodium Dichromate, Dihydrate, Technical
TT-M-261	Methyl Ethyl Ketone Technical
MMM-A-132	Adhesive, Heat Resistant, Airframe Structural, Metal to Metal

FSC 9330

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet

Military

MIL-Y-1140	Yarn, Cord, Sleeving, Cloth and Tape Glass
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-8949	Steel Bars, Plates, Sheets, Billets, and Reforging Stock Type D6AC
MIL-T-81533	Trichloroethane 1,1,1 (Methyl Chloroform Inhibited, Vapor Degreasing
MIL-A-83377	Adhesive Bonding for Aerospace Systems, Guidelines for

STANDARDS

Federal

Fed. Test Method Std. 151	Metal; Test Methods
Fed. Test Method Std. 406	Plastics: Methods of Testing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The materials furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2).

3.2 Type I. Type I material shall meet all of the requirements specified in Table I when cured for 60-70 minutes at 180-190°F plus 60-70 minutes at 350-360°F or cured to the manufacturers recommended step cure cycle.

3.3 Types II & III. Types II & III materials shall meet all of the requirements specified in Table I when cured for 120-130 minutes at 200-210°F or to the manufacturers recommended cure cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Quality conformance inspection

4.2.1 Acceptance tests.

(a) Lap shear strength at room temperature and 350°F for Type I and room temperature and 270°F for Types II & III. Metal shall be 2024-T81 clad.

(b) Ultimate compression strength at room temperature and at 350°F for Type I and at room temperature and 270°F for Types II & III.

(c) Density

4.2.2 Tests and test specimens shall be in accordance with this specification. Results shall meet the applicable requirement of Table I. Three test specimens shall be used for each property requirement specified in Table I for each type of material except that the shear strength specimens shall be in accordance with Figure 1 and 4.5.2.4. Specimens of each type shall be tested for each shipment or each batch if a shipment consists of more than 1 batch. Data from test shall be forwarded with the shipment.

4.2.3 Retests shall automatically be accomplished in the event of a test failure. Failure of the retest shall result in rejection of the represented material.

4.2.4 Verification of any test in Table I may be conducted at the option of the Purchaser on any material received on production orders. Failure of the compound to meet test requirements shall result in rejection of the lot and necessary action to correct the discrepancy.

4.3 Cleaning of Adherends

4.3.1 Aluminum (except where use of release agent is specified - no cleaning required in that case).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

TABLE I. Property requirements. (Page 1 of 2 pages)

PROPERTY	MATERIAL		TEST METHOD
	TYPE I	TYPES II & III	
Weight Loss, %, 350° 10 Hours, Max. Avg.	.5	.5	4.5.1
Shore D Hardness After Cure, Min. Avg.	95	93	Fed. Test Method Std. 406 Method 1083
Exposure Shrinkage, %, 350°F, 10 Hours, Max. Avg.	.2	.2	4.5.1
Cure Shrinkage, %, Max. Avg.	.35	.2	4.5.1
Density After Cure, lb/ft^3, Avg.	101	98	4.5.4
Lap Shear Fluid Resistance Strength, psi, Min. Avg.			4.5.2 4.5.3
(a) Salt Spray:			
Aluminum	1500	1500	
(b) Humidity:			
Aluminum	1700	1650	
(c) 180°F JP-4:			
Aluminum	1800	1850	
Minimum Average Compressive Strength, psi			4.5.4
(a) Ultimate at:			
-65°F	35,000	30,000	
Room Temp.	23,000	21,000	
270°F	12,500	10,000	
350°F	8,000	8,000	
420°F	5,500	----	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

TABLE I. Property requirements. (Page 2 of 2 pages)

PROPERTY	MATERIAL		TEST METHOD
	TYPE I	TYPES II & III	
(b) Proportional Limit at:			
-65°F	16,000	15,000	4.5.4
Room Temp.	10,000	9,000	
270°F	7,500	4,000	
350°F	6,000	----	
420°F	4,500	----	
(c) Minimum Modulus 10^5 psi at:			
-65°F	6.0	7.5	4.5.5
Room Temp.	5.0	6.5	
270°F	2.0	3.0	
350°F	1.1	1.1	
420°F	0.95	0.95	

4.3.1.1 Remove all surface dyes and other foreign material by solvent wiping with methyl-ethyl-ketone conforming to Federal Specification TT-M-261.

4.3.1.2 Vapor degrease parts with 1,1,1 Trichloroethane, inhibited conforming to Specification MIL-T-81533. The parts shall be suspended in vapor until condensation ceases.

4.3.1.3 The parts shall be immersed 9 to 15 minutes in a solution of the following composition which is maintained at a temperature of 145-160°F. (Solution composition shall be maintained within ±10%).

Water (tap)	30 parts by weight
Sulphuric Acid (66° Be) Fed. Spec. O-S-595	10 parts by weight
Sodium Dichromate Fed. Spec. O-S-595	4 parts by weight

4.3.1.4 Rinse the parts in clean tap water at room temperature for a minimum of 2 minutes, followed by a spray rinse for a minimum of two minutes in clean distilled or demineralized water at room temperature. If an air supply is used for spray rinsing, the air shall be filtered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

4.3.1.5 The panels shall be observed to make sure they hold a continuous film of water on the areas to be bonded. They shall be recleaned if the water film is not continuous.

4.3.1.6 The parts shall be dried in circulating air at 150°F or below.

4.3.1.7 Specimens shall be assembled and bonded within 72 hours of the completion of the cleaning process. Cleaned metal shall be protected from contamination during that time.

4.4 Curing of specimens

4.4.1 Specimens shall be cured in accordance with the applicable cure cycle as required by 3.2 or 3.3 or in accordance with the manufacturer's recommended cure cycle. When applying heat, the time from room temperature to the applicable cure temperature shall be not less than 10 minutes or more than 60 minutes.

4.4.2 All lap shear specimens and sandwich panel specimens shall be cured under 25 psi $\pm$5 psi pressure. One layer of Fabric 1165, MIL-Y-1140 glass fabric, plain weave shall be used as a bond line spacer in lap shear specimens. Bond line thickness shall be .005 to .010 for lap shear tests.

4.5 Specimen configuration and test procedure

4.5.1 Shrinkage and weight loss of molded specimens

4.5.1.1 A closed steel mold with a rectangular cavity approximately 16" x 1.5" x .5" is to be used to prepare specimens. Catalyzed material is to be injected into the mold until it is completely filled.

4.5.1.2 The material shall be cured using the specified time-temperature cycle, then cooled to approximately 75°F and removed from the mold.

4.5.1.3 The length of the specimen shall be determined to the nearest .001", and its weight to the nearest .01 gram.

4.5.1.4 The length of the mold cavity shall be determined to the nearest .001".

4.5.1.5 Shrinkage "as molded" is calculated by: % shrinkage

$$(\text{as molded}) = \left(\frac{\text{length of cavity - length as molded}}{\text{length as molded}}\right)$$

4.5.1.6 The specimen shall be exposed to the specified temperature condition (unrestrained).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

4.5.1.7 At the end of the exposure, the specimen shall be allowed to cool to room temperature, (70-80°F) and its weight and length determined to the previous accuracies.

4.5.1.8 Time required to raise specimen to exposure temperature shall not be included in exposure time.

4.5.1.9 % Weight loss at each test interval shall be calculated by:

$$\% \text{ weight loss} = \frac{\left(\begin{array}{c}\text{weight as} \\ \text{molded}\end{array} - \begin{array}{c}\text{weight after} \\ \text{exposure}\end{array}\right)}{\text{weight as molded}} \times 100$$

4.5.1.10 % Shrinkage after exposure shall be calculated by:

$$\% \text{ shrinkage} = \frac{\left(\begin{array}{c}\text{length as} \\ \text{molded}\end{array} - \begin{array}{c}\text{length after} \\ \text{exposure}\end{array}\right)}{\text{length as molded}} \times 100$$

4.5.2 Lap shear test - fluid exposure - See Figure 1.

4.5.2.1 Prepunched panels (finger panels) in accordance with MMM-A-132 or precut individual pieces shall be used for all lap shear tests.

4.5.2.2 Metal to be 2024-T81 clad (.064").

4.5.2.3 Specimens 1 and 6 are controls and shall be removed and held while exposure portion of panel is exposed to specified environment.

4.5.2.4 After proper exposure as specified in 4.5.3, specimens 2, 3, 4 and 5 shall be tested with controls, within 6 hours.

4.5.2.5 Load specimens with clamp type jaws using a loading rate of 600-700 lb/min.

4.5.2.6 Calculate shear in psi, τ, by $\tau = \frac{P}{LW}$

Where: P = failing load in lbs, L = overlap in inches, and
W = width of specimen in inches.

4.5.2.7 L and W shall be determined to the nearest .001".

4.5.3 Fluid resistance test exposure conditions

4.5.3.1 Salt water spray condition

4.5.3.1.1 Use Fed. Test Method Std. 151 Method 811.1 except as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

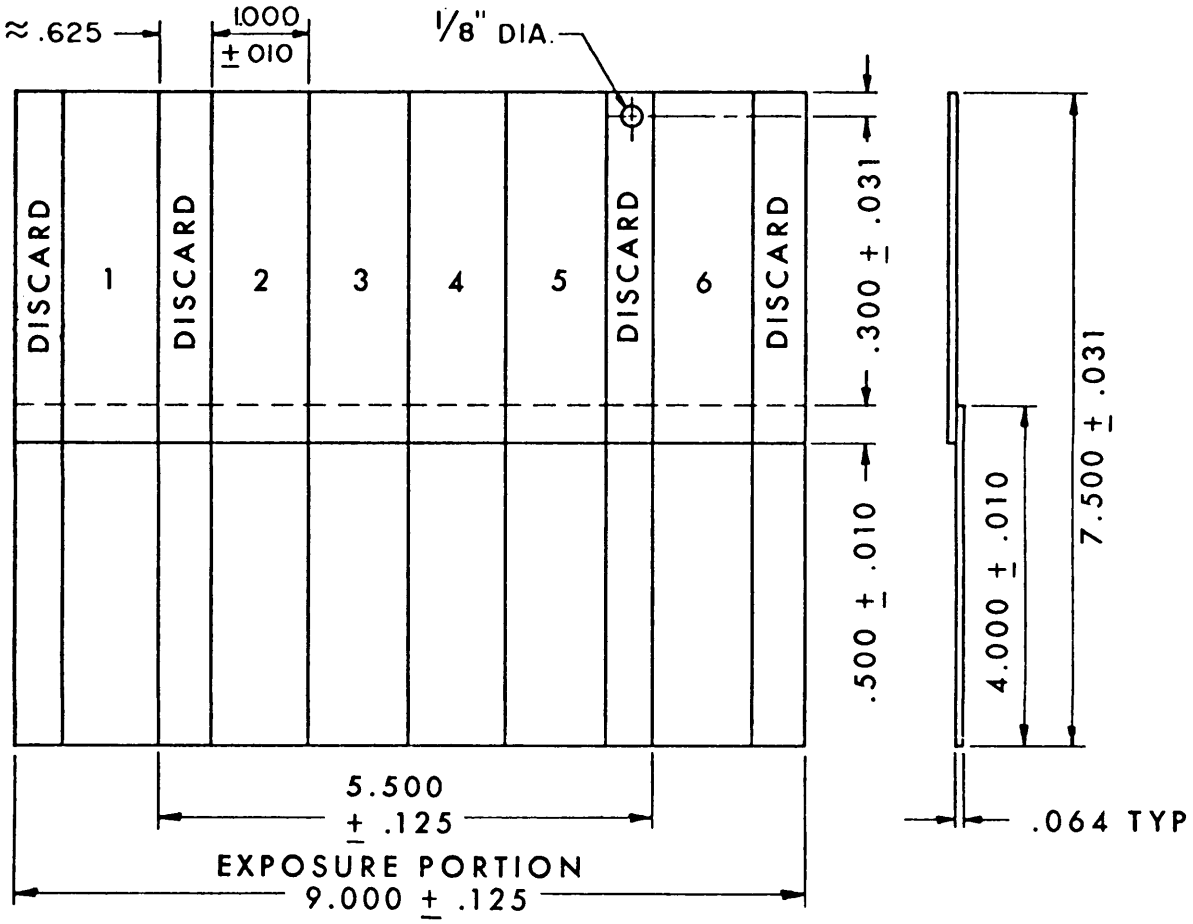


Figure 1

4.5.3.1.2 Suspend panel vertically by means of a non-metallic material, eg, waxed string, through a 1/8" diameter hole drilled in the trim area.

4.5.3.1.3 The plane of the panel is to be parallel to the principal direction of flow of the water spray.

4.5.3.1.4 Panels are not to be in contact with each other or any metallic object during exposure.

4.5.3.1.5 Exposure duration is to be 30 days ± 2 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

exposure means that the chamber shall be closed and the humidity maintained continuously except for brief interruptions necessary to inspect, rearrange, or remove panels, or to check or replenish the reservoir.

4.5.3.3 Fuel immersion condition

4.5.3.3.1 Panels shall be immersed in jet engine fuel (JP-4) per MIL-T-5624 maintained at 180°F ± 5°F for 7 days ± 2 hours.

4.5.3.3.2 Panels must be arranged so that the overlap area is fully exposed to the fuel.

4.5.3.3.3 The panels shall not come in contact with each other nor shall the bond line come into contact with any metallic material throughout the exposure.

4.5.4 Compression test and density - See Figure 2.

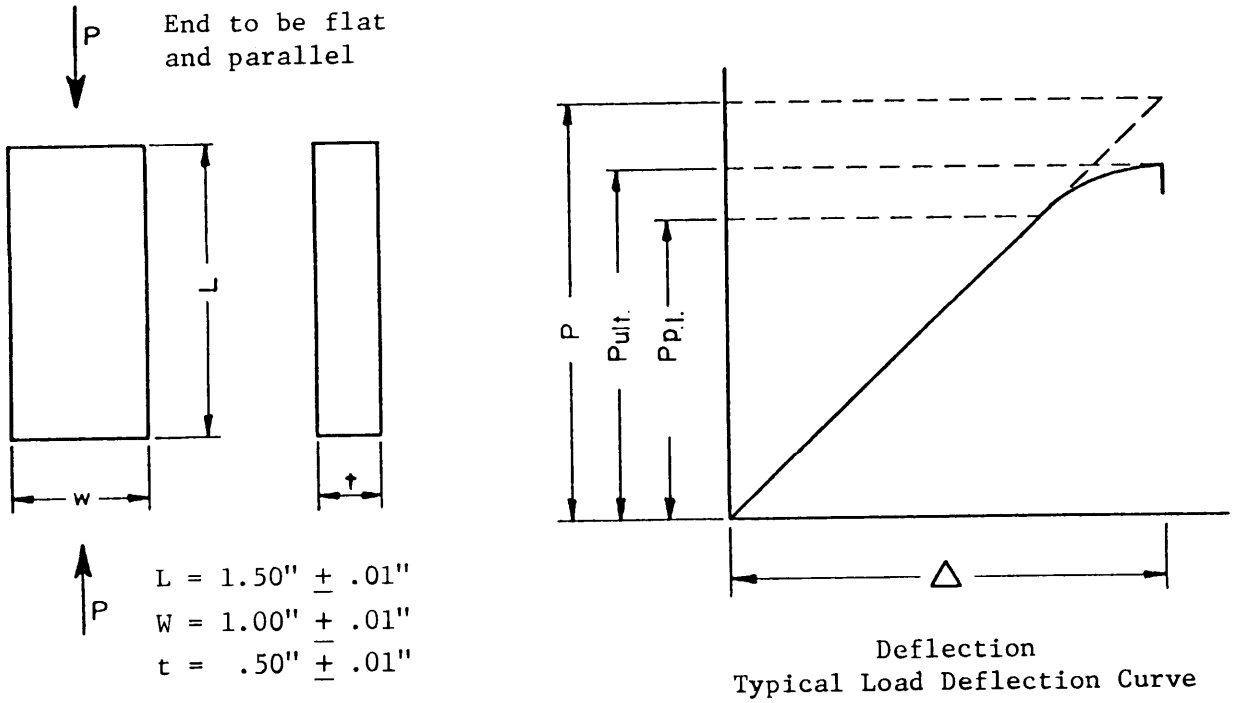
4.5.4.1 Prior to compression testing, density of each specimen shall be determined by weighing to the nearest .01 gram and converting the weight measurement to lbs/ft^3.

4.5.4.2 Load rate to be .05 inch per minute (head travel).

4.5.4.3 Specimens shall be molded and/or machined to tolerance as shown in Figure 2.

4.5.4.4 Load-deflection curve to be determined autographically.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.



NOTE: Test equipment shall have rigid non-rotating loading platforms. Specimen shall be unsupported, i.e. free to deflect laterally.

Figure 2

4.5.4.5 Calculate ultimate compressive strength σC_{ULT}, proportional limit compressive strength $\sigma C_{P.L.}$, and compression modulus E_c by

$$\sigma C_{ULT.} = \frac{P_{ULT.}}{tw} ; \quad \sigma C_{P.L.} = \frac{P_{P.L.}}{tw} ; \quad E_c = \frac{PL}{\Delta tw}$$

Where P_{ULT}, $P_{P.L.}$, P/Δ, L, t, and w are as shown in the above figures and formulas, $P_{P.L.}$ = proportional limit.

4.5.5 Compression creep resistance - See Figure 3.

4.5.5.1 Apply shim material to steel plate. Coat aluminum plate with release agent. Assemble plates as shown in Figure 3, but do not drill at this stage. Cure shim material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

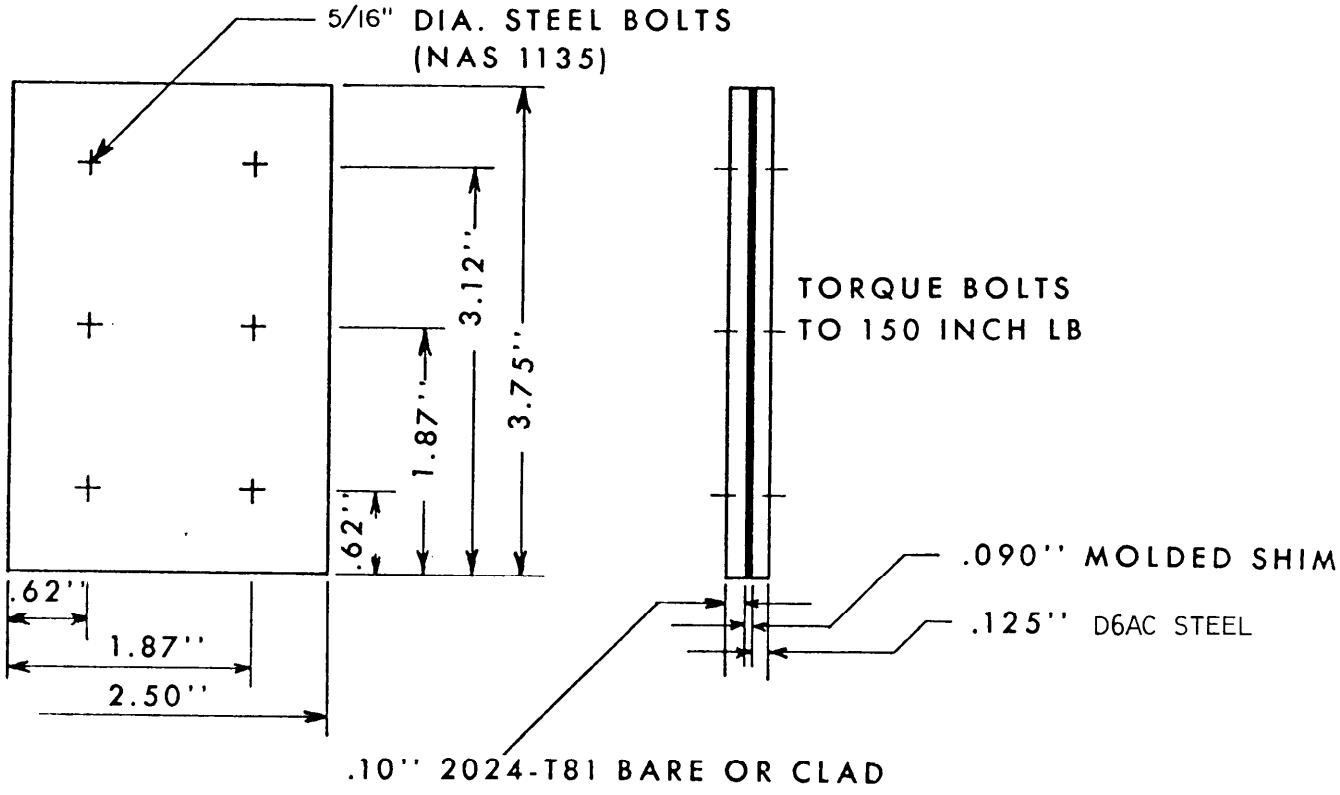


Figure 3

4.5.5.2 Drill holes as shown in Figure 3. Disassemble, deburr, reassemble and install bolts. Torque each bolt to 150 inch-lb. Identify bolts for reference.

4.5.5.3 Expose panel to temperature for total of 240 hours. Remove panel from oven and determine remaining torque in each bolt. Torque readings shall be taken after the specimen has cooled to 75°F $\pm$ 5°F by applying torque in the tightening direction and observing the torque value necessary to produce movement in the bolt. Determine % difference between initial torque and average of final torque of each bolt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation - packaging

5.1.1 Level C

5.1.1.1 Type I compound and curing agent shall be supplied in an air-tight, spill proof, water proof containers. The compound shall be procured in kit form consisting of a container of basic formulated resin and a container of its curing agent.

5.1.1.2 Type II compound shall be supplied in catalyzed sheet form frozen at -35°F or below. The size and thickness of the sheets shall be specified by the procuring agency.

5.1.1.3 Type III compound shall be provided in a frozen, precatalyzed condition (-35°F or below) packaged in a disposable container which is suitable for dispensing in a caulking gun.

5.1.2 Type I compound shall be stored at below 100°F.

5.2 Packing

5.2.1 Level C. The molded shim compound packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Each container shall be permanently and legibly marked with the following information.

Specification MIL-S-83474
Mfgrs. name and material designation
Type compound
Mfgrs. Lot _____ date of Mfgr
For Type I store below 100°F
For Type II store below -35°F.
For Type III store below -35°F.

5.4 Types II & III material shall be stored at -35°F or below.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The compounds covered by this specification are intended for use to form cured-in-place shims of sufficient structural strength for use during assembly of aircraft structural components and parts. The compounds shall not be used under the following conditions:

(a) Where the curing temperature will adversely affect the other materials which form a part of the installation.

(b) In any application which would subject the shim to a fretting condition.

6.2 Bondability. The molded shim material is bondable to other surfaces such as steels and titanium. Procedures as employed for structural adhesive bonding may be used. The user may modify these procedures to maximize the bond. AFML-TR-68-133, Fabrication Methods With Molded Shims For Aerodynamic Surface Control, contains some detailed surface preparation procedures.

Custodian:
Air Force - 11

Preparing Activity
Air Force - 11

Project No. 9330-F627

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.

FOLD

ASD/ ENESS
Wright-Patterson AFB, O 45433

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S.MAIL

ASD/ENESS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:15Z
Check the source to verify that this is the current version before use.